UNIT 07 TEACHER CORE LESSON • BIG IDEAS MATH (GRADES 5–6 • SC-6)

# Chained Operations: Evaluating (a + b) × (c - d) from the Inside Out

Numerical & Algebraic Expressions • Standard: CCSS.MATH.6.EE.A.2

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**1. MATHEMATICAL CONCEPT & STACK APPROACH: Evaluate multi-branch expressions from the inside out using prime factor towers and verify on the 4-register stack without parentheses.**

Written algebra uses parentheses to show which inner operations must happen first. Evaluating from the inside out follows true mathematical structure: compute the first branch (6 + 1 = 7), hold that result while computing the second branch (9 - 5 = 4), and multiply the two branch values together.

## 2. MODELING WITH BLOCKS & STUDENT PREDICTION

**Published Block Kit: Prime Factor Studio (Logarithmic Prime Blocks) & 4-Register Stack**

Hands-On Setup: Use Prime Factor Studio blocks in the 12-socket comparator lid to model the two factor branches: (6 + 1) = [7] prime block and (9 - 5) = [2] + [2] = 4 tower, then combine their heights for 7 × 4 = 28.

**Student Prediction (Before Calculator): Left sub-expression: 6 + 1 = 7. Right sub-expression: 9 - 5 = 4. Combined product: 7 × 4 = 28 (prime tower 2 × 2 × 7).**

## 3. CALCULATOR EXECUTION (SC-6 KEYCAPS & 4-LEVEL STACK TRACE)

**Keystroke Flow:** 6 ENTER 1 + 9 ENTER 5 - ×

| Key Pressed | Screen (X) | Internal 4-Level Stack Effect (T, Z, Y, X) |
|---|---|---|
| 6 | 6 | Value a (6) placed into Register X |
| ENTER | 6 | ENTER lifts 6 into Register Y |
| 1 | 1 | Value b (1) placed into Register X |
| + | 7 | Left operator (+): Combines Y(6) + X(1) = 7 in X |
| 9 | 9 | Value c (9) entered into X; 7 lifted to Y |
| ENTER | 9 | ENTER lifts 9 into Z and 7 into T |
| 5 | 5 | Value d (5) placed into Register X |
| - | 4 | Right operator (-): Combines Z/Y(9) - X(5) = 4 in X; 7 drops to Y |
| × | 28 | Root operator (×): Combines Y(7) × X(4) = 28 in X! |

**4. CONTEXT (Why 24 Is the Champion Packing Number & How Inside-Out Evaluation Eliminates Parentheses) • From Blocks to the Stack (SC-6 Manual) — Ch. 7: Prime Numbers & Factor Towers + Thinking in the Stack (SC-32 Manual) — Prologue & Ch. 2: Inside-Out Thinking**

Why are eggs sold by the dozen and hours divided into 24? Because 24 = 2 × 2 × 2 × 3 has eight divisors (1, 2, 3, 4, 6, 8, 12, 24), making it a champion packing number! And when evaluating nested expressions, 16th-century typesetters only invented parentheses to flatten 2D fraction bars onto a single line of type—whereas evaluating from the inside out resolves every branch naturally.

**Core Investigation (Student Warm-Up A):**

Evaluate (8 + 2) × (14 - 9) by finding the value of each parenthesized group first, then multiplying the two results.

**Teacher Answer Key: (8 + 2) = 10 and (14 - 9) = 5; 10 × 5 = 50 (Prime factors: 2 × 5 × 5).**

# Nested Fractions & Multi-Layer Expressions

Order of Operations & Nested Fractions • Standard: CCSS.MATH.6.EE.A.2

## 1. FOLLOW-UP EXTENSION CONCEPT, BLOCKS & PREDICTION

**Concept & Stack Approach: Deeply nested fractions such as (12 - 4) ÷ (2 × (1 + 1)) require evaluating the numerator (8) and the innermost group of the denominator (1 + 1 = 2, then 2 × 2 = 4) before dividing. The 4-register stack (T, Z, Y, X) mirrors this exact paper-and-pencil hierarchy.**

Blocks & Prediction: Model the numerator (12 - 4 = 8, three [2] blocks) and denominator (2 × (1 + 1) = 4, two [2] blocks) with Prime Factor towers. Prediction: Numerator: 12 - 4 = 8 (three [2] blocks). Denominator: 2 × 2 = 4 (two [2] blocks). Result: 8 ÷ 4 = 2.

## 2. EXTENSION CALCULATOR SEQUENCE & STACK TRACE

[1] [2] [ENTER] [4] [-] [2] [ENTER] [1] [ENTER] [1] [+] [×] [÷]

| Key Pressed | Screen (X) | Internal 4-Level Stack Effect |
|---|---|---|
| [1] | 1 | First digit of 12 |
| [2] | 12 | 12 in register X |
| [ENTER] | 12 | 12 lifted to Y |
| [4] | 4 | 4 in register X |
| [-] | 8 | Numerator 12 - 4 = 8 held in X |
| [2] | 2 | Factor 2 enters X; 8 lifts to Y |
| [ENTER] | 2 | 2 lifted to Y; 8 lifts to Z |
| [1] | 1 | Inner 1 enters X |
| [ENTER] | 1 | 1 lifted to Y; 2 in Z; 8 in T |
| [1] | 1 | Second 1 enters X |
| [+] | 2 | Innermost sum: 1 + 1 = 2 in X |
| [×] | 4 | Denominator combined: 2 × 2 = 4 in X; 8 in Y |
| [÷] | 2 | 8 / 4 = 2! |

## 3. TARGET MISCONCEPTIONS & HANDS-ON BLOCK REMEDIATIONS

**Misconception 1: Evaluating expressions strictly left-to-right (computing 10 + 5 × 2 as 15 × 2 = 30) instead of respecting operator structure.**

Hands-On Block Remediation: Model the product branch 5 × 2 with Prime Factor [5] and [2] blocks first: push 10 to Register Y ([10] [ENTER]), evaluate 5 × 2 in X ([5] [ENTER] [2] [×] → 10), then press [+] → 20.

**Misconception 2: Losing track of intermediate results when evaluating two parenthesized branches like (8 + 2) × (9 − 4).**

Hands-On Block Remediation: Show how Register Y holds the first branch (8 + 2 = 10) automatically while the second branch (9 − 4 = 5) is computed in X, ready for [×] = 50.

**Extension Challenge (Student Warm-Up B):**

Evaluate (15 - 5) ÷ (3 + 2) by computing the numerator and denominator separately, then dividing.

**My Paper Answer:** _____ **[ ] Verified on SC-6 (Online Key: stackcalc.io/curriculum/answers)**

STUDENT LAB WORKSHEET • UNIT 07 (SC-6)

# Postfix Evaluation, Factor Trees & Chained Operations

Name: _____ Date: _____ Class: _____

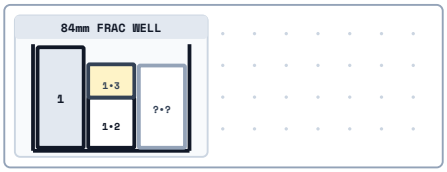
### STUDENT REFERENCE GUIDE: INSIDE-OUT EVALUATION & PRIME FACTOR TREE RULES

- Order of operations works from the inside out: evaluate inner groups, parentheses, and factor branches before combining them.
- Every composite number decomposes into a unique tower of prime factors (e.g., $28 = 2 \times 2 \times 7$).
- In nested fractions such as $(a - b) \div (c + d)$, the fraction bar groups the numerator and denominator as separate sub-problems.

**Key Sequence:** `6` `ENTER` `1` `+` `9` `ENTER` `5` `-` `×`

**Investigation 1: Evaluate (4 + 3) × (10 - 6) by solving each parenthesized group first.**

**1. Factor Tree / Grouping Sketch:**




**2. Inside-Out Evaluation Steps:**

Step 1 (Inner Group / Power):

Step 2 (Next Operation):

Final Value / Prime Factors:

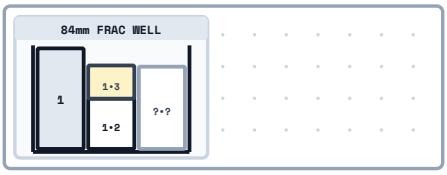
**3. My Prediction:**

**Final Verified Answer:**

[ ] Checked on SC-6

**Investigation 2: Evaluate (20 - 4) ÷ (2 × 4) by finding the numerator and denominator first.**

**1. Factor Tree / Grouping Sketch:**




**2. Inside-Out Evaluation Steps:**

Step 1 (Inner Group / Power):

Step 2 (Next Operation):

Final Value / Prime Factors:

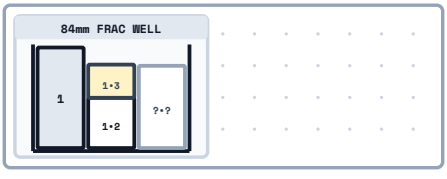
**3. My Prediction:**

**Final Verified Answer:**

[ ] Checked on SC-6

**Investigation 3: Evaluate (18 - 6) ÷ (1 + 2) and write the prime factorization of the numerator.**

**1. Factor Tree / Grouping Sketch:**




**2. Inside-Out Evaluation Steps:**

Step 1 (Inner Group / Power):

Step 2 (Next Operation):

Final Value / Prime Factors:

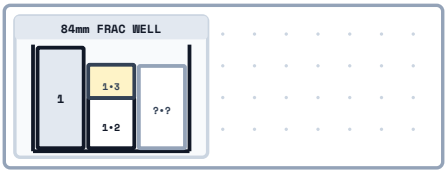
**3. My Prediction:**

**Final Verified Answer:**

[ ] Checked on SC-6

**Investigation 4: Evaluate 5 + (3 × 4) by computing the inner product before adding 5.**

**1. Factor Tree / Grouping Sketch:**




**2. Inside-Out Evaluation Steps:**

Step 1 (Inner Group / Power):

Step 2 (Next Operation):

Final Value / Prime Factors:

**3. My Prediction:**

**Final Verified Answer:**

[ ] Checked on SC-6

UNIT 07 PRACTICE WORKSHEET A • GRADES 5–6 (SC-6)

# Unit 07 Practice A: Core Fluency

Name: _____ Date: _____ Score: _____ / 10

**TIER 1 • LEVEL 1: BUILDER FOUNDATION • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE A (#1–#10) ON PAPER FIRST**

**EXAMPLE 1** **Find the prime factorization of 36.**

$36 = 4 \times 9 = 2^2 \times 3^2$. Key: [36] [FACT].

Verified Answer: $2^2 \times 3^2$

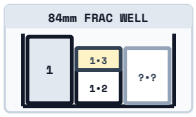
**EXAMPLE 2** **Find GCF(18, 24) and LCM(18, 24).**

$18 = 2 \times 3^2$, $24 = 2^3 \times 3$ → Shared GCF $= 2 \times 3 = 6$; Union LCM $= 2^3 \times 3^2 = 72$.

Verified Answer: GCF=6, LCM=72

**1** Tier 1 • Builder

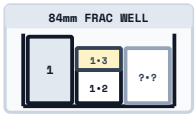
**Evaluate the power: $2^5$ =**




Paper Step: _____ [ ] SC-6 Ans: _____

**2** Tier 1 • Builder

**Evaluate the power: $3^4$ =**




Paper Step: _____ [ ] SC-6 Ans: _____

**3** Tier 1 • Builder

**Write the prime factorization of 24 using exponents:**




Paper Step: _____ [ ] SC-6 Ans: _____

**4** Tier 1 • Builder

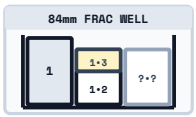
**Write the prime factorization of 60 using exponents:**




Paper Step: _____ [ ] SC-6 Ans: _____

**5** Tier 1 • Builder

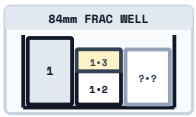
**Is 29 a prime number or a composite number?**




Paper Step: _____ [ ] SC-6 Ans: _____

**6** Tier 1 • Builder

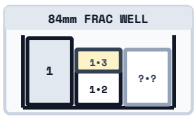
**Find the Greatest Common Factor: GCF(16, 24) =**




Paper Step: _____ [ ] SC-6 Ans: _____

**7** Tier 1 • Builder

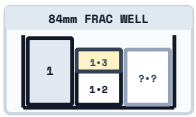
**Find the Greatest Common Factor: GCF(30, 45) =**




Paper Step: _____ [ ] SC-6 Ans: _____

**8** Tier 1 • Builder

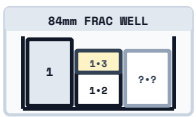
**Find the Least Common Multiple: LCM(6, 8) =**




Paper Step: _____ [ ] SC-6 Ans: _____

**9** Tier 1 • Builder

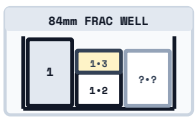
**Find the Least Common Multiple: LCM(12, 15) =**




Paper Step: _____ [ ] SC-6 Ans: _____

**10** Tier 1 • Builder

**Use GCF × LCM = $a \times b$: If GCF(12, 18) = 6, find LCM(12, 18):**




Paper Step: _____ [ ] SC-6 Ans: _____